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AN ADAPTIVE FUZZY CONTROL SYSTEM FOR METAHEURISTIC CONFIGURATION THROUGH DYNAMIC SELECTION OF FUZZY SCHEMES

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RESUMEN

Los algoritmos metaheurísticos son ampliamente utilizados para resolver problemas complejos de optimización en diversos dominios, debido a su flexibilidad y eficiencia. Sin embargo, su rendimiento depende críticamente de la configuración adecuada de los parámetros internos y de las estrategias de búsqueda. Las configuraciones estáticas o específicas para un problema a menudo no logran adaptarse a la naturaleza dinámica del proceso de búsqueda. Para abordar esta limitación, esta tesis propone un sistema de control difuso adaptativo para metaheurísticas, que configura dinámicamente sus parámetros mediante la selección, durante la ejecución, de esquemas difusos predefinidos.

El sistema integra lógica difusa en el ciclo de control del proceso de optimización, permitiendo flexibilidad semántica y plasticidad comportamental. Cada esquema difuso representa un conjunto coherente de funciones de membresía con características específicas, como forma, solapamiento y granularidad, que proporcionan interpretaciones semánticas alternativas de las variables de entrada. El controlador propuesto monitorea indicadores en tiempo real del comportamiento del proceso de optimización—como estancamiento, tasa de mejora y diversidad poblacional—para cambiar dinámicamente entre esquemas difusos. Esto introduce una forma novedosa de adaptación semántica en sistemas de control difuso.

La arquitectura está diseñada para admitir múltiples estrategias de selección dinámica, incluidas técnicas estocásticas, aprendizaje por refuerzo e hiperheurísticas. El sistema propuesto se integra en metaheurísticas de última generación, específicamente *Particle Swarm Optimization* (PSO) y *Genetic Algorithm* (GA), y se valida en problemas de referencia de dominios continuos, combinatorios y de selección de características.

Esta investigación aporta una metodología generalizable para el control adaptativo de parámetros en metaheurísticas, ofreciendo mejoras en desempeño, robustez y capacidad de adaptación. Se espera que los resultados proporcionen nuevas perspectivas sobre el rol de la dinámica semántica en controladores basados en lógica difusa y su impacto en la optimización metaheurística.

Palabras clave: Optimización, Metaheurísticas, Control de Parámetros, Lógica Difusa, Sistema de Control Difuso, Adaptación Semántica, Selección Dinámica.

ABSTRACT

Metaheuristic algorithms are widely employed to solve complex optimization problems in diverse domains due to their flexibility and efficiency. However, their performance critically depends on the proper configuration of internal parameters and search strategies. Static or problem-specific configurations often fail to adapt to the dynamic nature of the search process. To address this limitation, this thesis proposes an adaptive fuzzy control system for metaheuristics, which dynamically configures their parameters by selecting among predefined fuzzy schemes during execution.

The system integrates fuzzy logic into the control loop of the optimization process, enabling semantic flexibility and behavioral plasticity. Each fuzzy scheme represents a coherent set of membership functions with specific characteristics, such as shape, overlap, and granularity, providing alternative semantic interpretations of the input variables. The proposed controller monitors real-time indicators of the optimization behavior—such as stagnation, improvement rate, and population diversity—to dynamically switch between fuzzy schemes. This introduces a novel form of semantic adaptation in fuzzy control systems.

The architecture is designed to support multiple dynamic selection strategies, including stochastic methods, reinforcement learning, and hyperheuristics. The adaptive fuzzy control system is integrated into state-of-the-art metaheuristics, specifically Particle Swarm Optimization (PSO) and Genetic Algorithm (GA), and validated on benchmark problems from continuous, combinatorial, and feature selection domains.

This research contributes a generalizable methodology for adaptive parameter control in metaheuristics, offering improved performance, robustness, and adaptability. The results are expected to provide insights into the role of semantic dynamics in fuzzy logic-based controllers and their impact on metaheuristic optimization.

Keywords: Optimization, Metaheuristics, Parameter Control, Fuzzy Logic, Fuzzy Control System, Semantic Adaptation, Dynamic Selection.

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1. INTRODUCTION

Metaheuristic algorithms have proven to be powerful tools for solving complex optimization problems, especially when exact methods become computationally intractable (Agung & Christine, 2021; Duan et al., 2023; Jaber et al., 2024). However, their performance strongly depends on the appropriate configuration of internal parameters and mechanisms (Dorigo et al., 1996; Hamadi et al., 2010; Kennedy & Eberhart, 1995). Achieving a balance between exploration (searching new regions) and exploitation (refining known promising areas) often requires configuration strategies that are either problem-specific or computationally expensive (Cáceres et al., 2017; Huang et al., 2019). In this context, adaptive mechanisms that adjust metaheuristic behavior during execution (based on real-time feedback) have gained increasing attention for their potential to improve solution quality while reducing computational cost (Eftimov et al., 2020; Z.-C. Lin, 2020).

Fuzzy logic provides a mathematical framework for modeling imprecise information and reasoning under uncertainty, making it particularly suitable for problems where crisp thresholds and rigid rules are insufficient (Zadeh, 1992; Zadeh, 1983). Fuzzy control systems, especially those based on the Mamdani model, have been successfully applied to dynamically adjust parameters in metaheuristic algorithms (Díaz-Cortés et al., 2017; Valdez et al., 2014). By encoding expert knowledge or behavioral patterns through linguistic rules and membership functions, fuzzy controllers allow for nuanced, real-time adjustments that improve the balance between exploration and exploitation (Sadollah, 2018; Valdez et al., 2010). This adaptability has shown promising results in enhancing the performance of metaheuristics across a variety of problem domains (Sadollah, 2018; Valdez et al., 2014).

This proposal aims to develop and validate an adaptive fuzzy control system for the dynamic configuration of metaheuristics, focusing on the selection of fuzzy schemes during the execution of the optimization process. This mechanism is designed to improve the system's capacity to adapt to changing search conditions inherent to the stochastic nature of metaheuristics, thereby enabling a flexible semantic representation of the fuzzy controller's input variables, and allowing for semantic flexibility and behavioral plasticity as the system modifies its interpretation of the optimization landscape in response to real-time indicators (Dombi & Csiszár, 2021; Valdez et al., 2014).

Unlike traditional approaches that rely on fixed membership functions or static fuzzy controllers, this research proposes an adaptive architecture capable of dynamically selecting among a set of predefined fuzzy schemes. Each profile defines a specific fuzzy partition of the domain of a linguistic variable, consisting of a coherent set of membership functions that may differ in type, quantity, shape, distribution over the domain, and degree of overlap (Pedrycz & Wang, 2015; Sadollah, 2018). These differences allow the system to represent alternative semantic interpretations of input variables. The selection of an appropriate fuzzy profile during runtime is guided by the observed behavior of the optimization algorithm, such as stagnation, relative improvement, or diversity loss (Castillo et al., 2017; Patel & Shah, 2022b), thereby introducing

a novel form of semantic adaptation within fuzzy control systems. This capability is positioned as a methodological innovation in the dynamic configuration of metaheuristics regulating parameter control behaviour, where the output of the fuzzy system corresponds to the adaptation of metaheuristics parameters (Díaz-Cortés et al., 2017; Valdez et al., 2014).

The dynamic selection mechanism of fuzzy schemes is not predetermined in this proposal but rather defined as a core research objective. The project will explore, implement, and comparatively evaluate several dynamic selection strategies, including static mechanisms (Cord et al., 2001), stochastic approaches, reinforcement learning techniques, and hyperheuristic methods (Burke et al., 2010; Crawford et al., 2012; Haighton et al., 2024; Qiu et al., 2020; Zhu et al., 2020). The performance of these strategies will be empirically assessed to determine which is most effective in the context of adaptive metaheuristic configuration for optimization problems.

The adaptive fuzzy control system will be integrated into leading state-of-the-art metaheuristics, such as Particle Swarm Optimization (PSO) and Genetic Algorithm (GA) (Valdez et al., 2014), and validated through their application to classical benchmark problems, including mathematical functions from the IEEE Congress on Evolutionary Computation (CEC), combinatorial problems such as the Set Covering Problem (SCP) and the Knapsack Problem (KP) (Chatterjee et al., 2024; Fatih Tasgetiren et al., 2006), and the Feature Selection Problem (Sadollah, 2018). A comparative experimental evaluation will be conducted to assess the performance of the different dynamic selection strategies under controlled conditions. This validation will allow for a detailed analysis of the impact of each mechanism on the effectiveness, efficiency, and robustness of the optimized algorithms (Crawford et al., 2021; Djebedjian et al., 2021; Halim et al., 2021). The expected outcome is a validated adaptive fuzzy control system capable of improving the performance of metaheuristic algorithms across diverse problem types.

The remainder of this document is organized into four main chapters. Chapter 2 presents the theoretical foundations of the proposal, including the fundamentals of optimization, metaheuristics, fuzzy logic, fuzzy control systems, and the proposed semantic adaptation mechanism via fuzzy schemes. Chapter 3 describes the research proposal in detail, outlining the motivation, design principles, functional architecture, and validation strategy for the adaptive fuzzy control system. Chapter 4 defines the methodological framework, including the research questions, hypothesis, and general objective guiding the investigation. Finally, Chapter 5 presents the work plan, structured into work packages with defined tasks, expected outputs, and a Gantt chart that delineates the timeline for the implementation and evaluation of the proposed system.

2. THEORETICAL FRAMEWORK

2.1. Optimization and Metaheuristics

2.1.1. Optimization Problems

An optimization problem in computer science involves identifying the best possible solution, typically a maximum or minimum, of an objective function, subject to a set of constraints imposed on the decision variables. As stated by Houssein et al. (2024), optimization addresses the fundamental question: *how to obtain the best solution for a problem while considering a set of limitations?* In their recent work, Houssein et al. (Houssein et al., 2024) also propose a comprehensive taxonomy that classifies optimization problems according to the nature of decision variables (discrete, continuous, or mixed), the presence or absence of constraints, and structural characteristics such as multi-objectivity and combinatorial complexity. This framework reflects typical configurations encountered in real-world engineering and computational applications.

Figure 2.1 offers a visual summary of this taxonomy, illustrating the interrelationships among different problem classes and highlighting their relevance in practice.

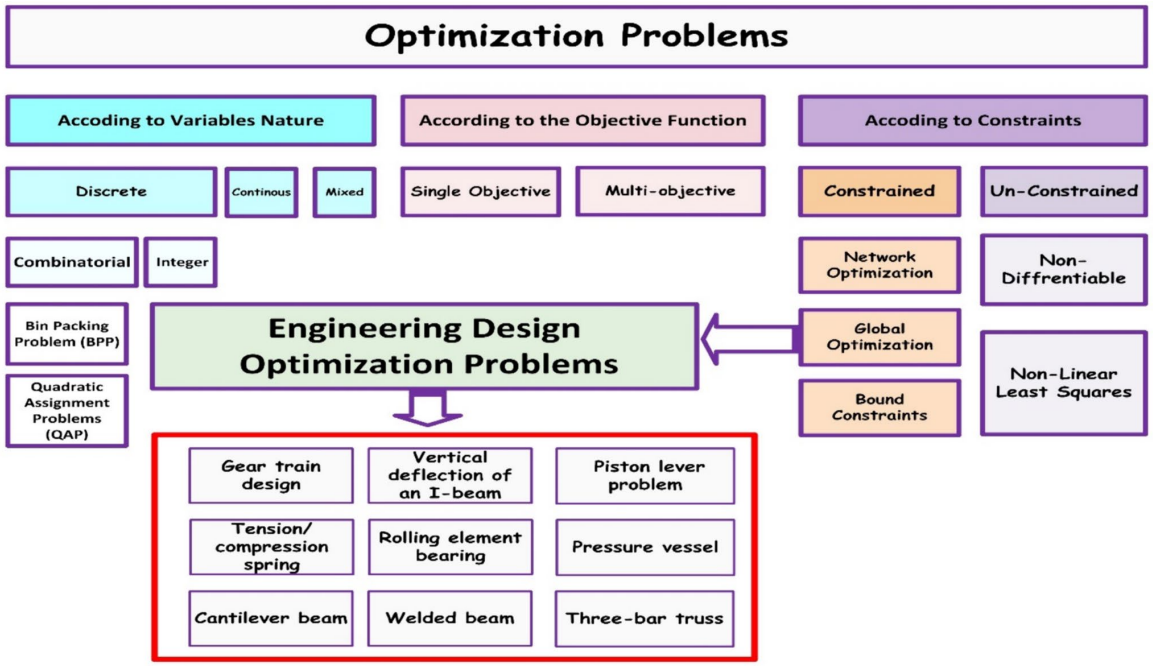


Figure 2.1: Taxonomy of optimization problems based on decision variables, objective function type, and constraints. (Houssein et al., 2024).

A generic optimization problem is composed of several key components. At its core is the *objective function*, which is a mathematical expression to be maximized or minimized, typically representing cost, performance, utility, or another measure of interest (Carissimo & Korecki,

2024; Khomenko, 2018). The variables over which this function is defined are known as *decision variables*; these represent the unknown quantities that must be determined in order to optimize the objective (Khomenko, 2018; Zhan et al., 2022; Zheng & Liu, 2022).

Constraints are conditions imposed on the decision variables, typically formulated as equalities or inequalities, which define the subset of solutions that are considered feasible (Zhan et al., 2022; Zheng & Liu, 2022). The set of all feasible solutions forms the *search space*, representing all possible combinations of variable values that satisfy the given constraints (Khomenko, 2018). Finally, the *optimal solution* is defined as the feasible configuration that produces the best possible value—either a maximum or a minimum—of the objective function (Carissimo & Korecki, 2024).

To consolidate the previous definitions, Algorithm 2.1 formalizes the general structure of an optimization problem using standard mathematical notation. Let $X = \{x_1, x_2, \ldots, x_n\}$ denote the set of decision variables, where each x_i belongs to a domain of permissible values. The objective function $f : \mathbb{R}^n \to \mathbb{R}$ maps each candidate solution $x \in \mathbb{R}^n$ to a real-valued performance score. A set of constraints $\mathcal{C} = \{c_1(x), c_2(x), \ldots, c_m(x)\}$ defines the feasibility of each solution by restricting the values that x can assume.

The feasible region, denoted by $\mathcal{S}$, is the subset of $\mathbb{R}^n$ consisting of all points that satisfy every constraint c_i. The optimization task is to find a solution $x^* \in \mathcal{S}$ such that the value $f(x^*)$ is optimal—either minimal or maximal—depending on the problem formulation.

Algoritmo 2.1 Formal Structure of an Optimization Problem

Require: Set of decision variables $X = \{x_1, x_2, \ldots, x_n\}$
Require: Objective function $f : \mathbb{R}^n \to \mathbb{R}$
Require: Set of constraints $\mathcal{C} = \{c_1(x), c_2(x), \ldots, c_m(x)\}$
1: Define feasible region: $\mathcal{S} \leftarrow \{x \in \mathbb{R}^n \mid c_i(x) \text{ is satisfied } \forall c_i \in \mathcal{C}\}$
2: Solve:

$$x^* = \arg\min_{x \in \mathcal{S}} f(x) \quad \text{or} \quad x^* = \arg\max_{x \in \mathcal{S}} f(x)$$

3: **return** Optimal solution x^* and optimal value $f(x^*)$

This formal structure provides a unified framework for modeling a wide variety of optimization problems. However, depending on the nature of the decision variables and the characteristics of the search space, these problems can be categorized into distinct types. In particular, the distinction between continuous and discrete variables plays a crucial role in determining the appropriate solution techniques, as it directly influences algorithmic design, computational complexity, and the selection of metaheuristic or exact methods.

Among the different categories identified in the taxonomy, the distinction between continuous and discrete optimization problems is particularly relevant for algorithm design.

In *continuous optimization problems*, the decision variables can assume any real value within bounded or unbounded intervals. These problems often feature smooth and differentiable

objective functions and are typically addressed using gradient-based or calculus-driven techniques (Chen et al., 2009; Stein et al., 2004). In contrast, *discrete optimization problems* restrict the decision variables to finite or countable sets, such as integers or categorical values. The resulting search space consists of isolated points, which precludes the use of differential calculus and requires specialized combinatorial or enumerative strategies (Bartz-Beielstein & Zaefferer, 2017; Shen & Yu, 2018).

A particularly challenging subset of discrete optimization is *combinatorial optimization*, where the objective is to select the best configuration or subset of elements under structural or logical constraints. Problems in this class, such as the traveling salesman problem (TSP) and the knapsack problem (KP), are notorious for their exponential growth in solution space as the problem size increases (Fatih Tasgetiren et al., 2006; Santucci et al., 2020; Zhao & Gu, 2024). Many of these problems are classified as NP-hard, meaning that no known algorithm can solve all instances efficiently in polynomial time. In such cases, finding or even verifying optimal solutions may require exponential time in the worst case (Ahmed et al., 2021; Buchheim & Kurtz, 2018). Classic NP-hard examples include TSP, KP, the maximum cut problem, and the vertex cover problem (Chatterjee et al., 2024; Schuetz et al., 2022), all of which exhibit severe scalability issues as dimensionality increases (Dutta et al., 2021; Honari-Latifpour et al., 2022).

Due to these limitations, a wide range of approximation algorithms, heuristics, and metaheuristics has emerged as an alternative to exact methods. These techniques aim to provide high-quality, near-optimal solutions within acceptable computational budgets and time constraints. Among them, metaheuristics stand out for their flexibility, robustness, and ability to adapt to large-scale, high-dimensional, or dynamically changing problem landscapes (Chatterjee et al., 2024; Q. Wang & Tang, 2021).

While exact approaches such as linear programming (LP), mixed-integer programming (MIP), and branch-and-bound remain important tools for small or well-structured problems, their practical application is often constrained by several factors. These include poor scalability—as the computational cost increases rapidly with problem size (Agung & Christine, 2021; Duan et al., 2023)—as well as time sensitivity, where the need for rapid decision-making surpasses the response capacity of exact solvers (Jaber et al., 2024). Moreover, these methods generally rely on static, fully specified models and therefore perform poorly under conditions of uncertainty, dynamic constraints, or evolving objectives (Achamrah, 2024).

As a result, heuristic and hybrid strategies have gained increasing attention in both theoretical and applied optimization. Among them, metaheuristic algorithms offer a promising and versatile framework for addressing complex optimization problems, particularly in combinatorial and NP-hard domains. Their adaptability, scalability, and independence from gradient information make them especially suited to the types of problems addressed in this research (Talbi, 2009).

This context justifies the adoption of adaptive metaheuristic strategies—capable of dynamically responding to problem complexity—which are explored in the following section.

2.1.2. Metaheuristics: Principles and Taxonomy

Metaheuristics have become a central paradigm in the field of optimization, particularly within the broader category of approximate or incomplete solution methods. Unlike exact algorithms, which offer guarantees of optimality at high computational cost, metaheuristics provide high-quality solutions with reasonable effort, making them suitable for hard optimization problems (Gendreau, Potvin, et al., 2010; Sörensen et al., 2018; Talbi, 2009). Their applicability spans combinatorial, continuous, discrete, and mixed-variable domains, positioning them among the most widely adopted optimization strategies in both academic and industrial settings.

By abstracting away problem-specific details and relying on general exploration and exploitation strategies, metaheuristics offer modularity, flexibility, and adaptability. Their effectiveness stems from customizable components—such as variation operators, selection mechanisms, encoding schemes, and control parameters—that can be tuned or adapted to match different problem landscapes (Glover, 1986; Tzanetos et al., 2020).

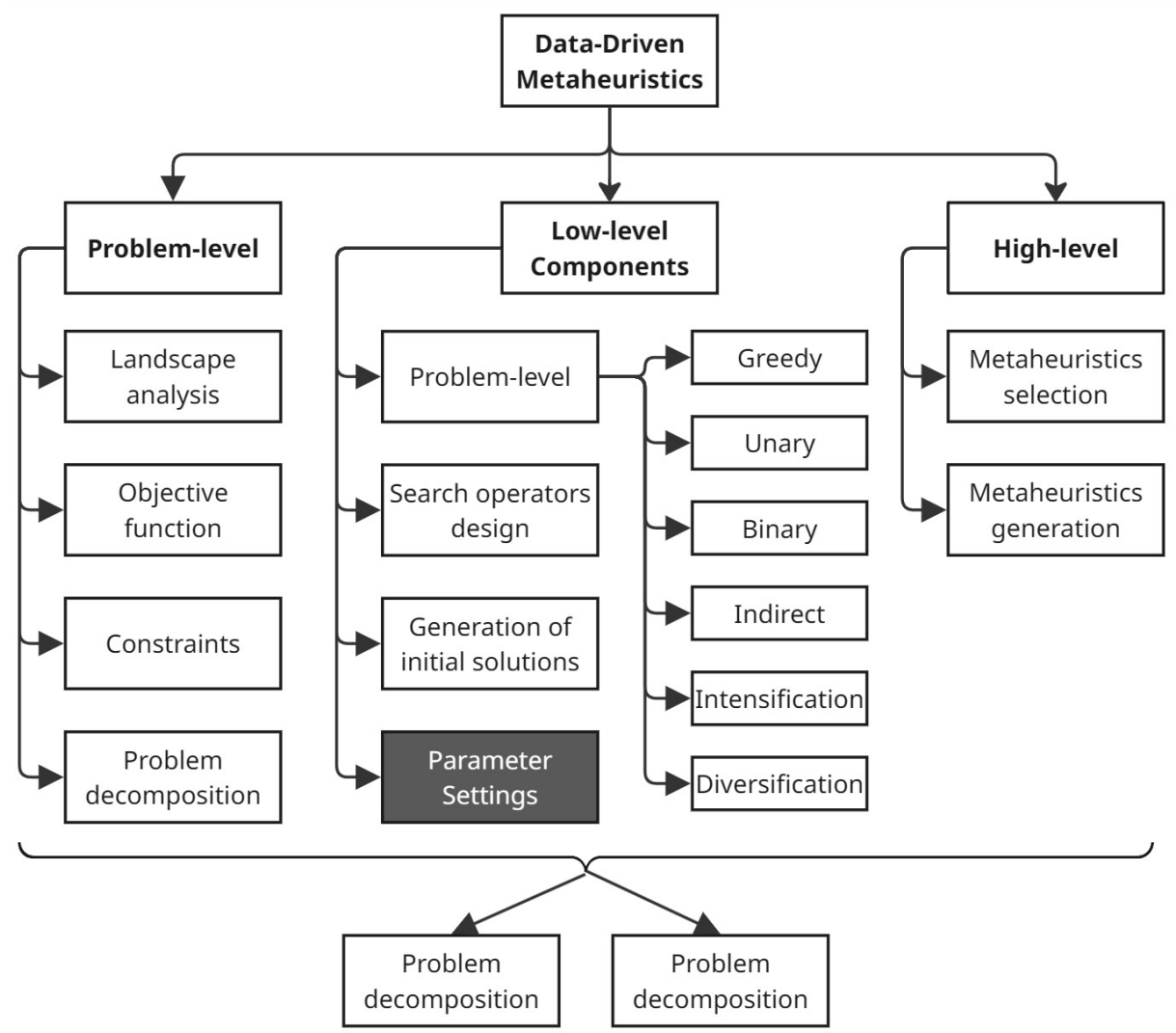


Figure 2.2: A general taxonomy of data-driven metaheuristics. Adapted from Talbi

A major research direction in recent years has been the development of adaptive and data-driven metaheuristics. These approaches aim to reduce reliance on expert configuration by incorporating learning mechanisms that allow the algorithm to adapt based on performance feedback. Talbi (Z.-C. Lin, 2020) proposed a hierarchical taxonomy of data-driven metaheuristics, which distinguishes three levels of integration: (1) the problem level (e.g., landscape analysis or problem decomposition), (2) low-level components (e.g., adaptive operators), and (3) high-level strategy control (e.g., algorithm selection based on learned patterns).

A closely related dimension of adaptability is parameter configuration. The configuration of metaheuristic parameters plays a pivotal role in determining algorithm behavior, convergence dynamics, and overall performance. Parameters govern how the algorithm explores and exploits the search space. Poor configurations may lead to premature convergence or inefficient behavior. Three main approaches have been studied: static tuning, which relies on offline configuration through expert knowledge or trial-and-error (Dorigo et al., 1996; Hamadi et al., 2010; Kennedy & Eberhart, 1995); dynamic parameter control, which adjusts values at runtime in response to feedback signals (Cáceres et al., 2017; Huang et al., 2019); and self-adaptive mechanisms, where parameters evolve along with candidate solutions. Recent trends highlight the emergence of autonomous search frameworks (Karaboga & Basturk, 2007), aiming to embed intelligent, runtime adaptation into the algorithm itself. These developments provide a conceptual foundation for integrating fuzzy logic controllers as interpretable and flexible tools for dynamic parameter adaptation.

Positioning of this Work within Low-Level Online Adaptation

Particularly relevant to this proposal is the Parameter Settings component at the Low-level layer. Traditionally, this task required expert-defined rules or static parameter tuning, but data-driven methods now offer the possibility of controlling them through feedback mechanisms. In Talbi's taxonomy, such mechanisms can operate in offline mode (before execution) or online, meaning during runtime, enabling dynamic responses to observed search dynamics. Our proposal is situated within the low-level component as an online parameter setting, where we aim to control metaheuristic parameters based on search behavioral metrics such as stagnation, improvement rate, diversity, or convergence speed through the iterations. In particular, the system will be integrated into Particle Swarm Optimization (PSO) and Genetic Algorithm (GA), as representative and widely used metaheuristics.

Recent advances show that fuzzy logic-based controllers are particularly suitable for dynamic parameter adaptation. These systems use fuzzy rules to infer parameter adjustments from linguistic descriptors of algorithm behavior. For example, if diversity is low and improvement stagnates, a fuzzy rule may increase the exploration tendency by adjusting parameters accordingly. This strategy allows real-time control over the exploration–exploitation balance using interpretable and flexible rule sets. As such, fuzzy controllers offer a promising avenue for embedding semantic adaptability in metaheuristic configuration, using the data-driven paradigm described above (Eftimov et al., 2020). Our data-driven proposal: a modular fuzzy control system capable of selecting predefined fuzzy schemes during the execution of a metaheuristic al-

gorithm. These profiles encode different configurations of membership functions that influence how the controller interprets the state of the search through well-defined process metrics, and use this information to regulate the internal parameters of the algorithm dynamically. Through this adaptive mechanism, we aim to improve metaheuristic robustness, responsiveness, and performance in solving optimization problems.

Given the limitations of static or purely self-adaptive strategies for parameter control, fuzzy logic has emerged as a compelling alternative. It is capable of modeling uncertain, imprecise, or nonlinear relationships through interpretable linguistic rules. Its ability to encode expert knowledge and support real-time decision-making makes it particularly suitable for embedding adaptive behavior into metaheuristics.

In what follows, we present the theoretical and mathematical foundations of fuzzy logic and fuzzy control systems, which underpin the semantic-level adaptivity at the core of our proposed architecture.

2.2. Fuzzy Logic: Foundations

2.2.1. Fuzzy Set Theory and Membership Functions

Fuzzy logic is a mathematical framework developed to model and reason with vague, imprecise, or incomplete information, particularly in scenarios where traditional binary logic fails to capture the complexity of real-world conditions. Introduced by Zadeh (Zadeh, 1992; Zadeh, 1983), fuzzy logic enables inference with degrees of truth, where propositions are not constrained to being absolutely true or false, but rather can hold partial truth values ranging continuously between 0 and 1.

This capacity to handle gradation in truth values allows fuzzy logic to effectively represent linguistic concepts such as "hot", "fast", or "high", which are inherently ambiguous. For example, a person may be considered 0.7 tall and 0.3 not tall, depending on the selected membership function. Classical logic, in contrast, enforces a binary classification of this property, which limits its expressive power and its capacity to manage uncertainty (Sabahi & Akbarzadeh-T, 2013; Yen, 1999).

Fuzzy logic is grounded in fuzzy set theory, where each element x in a universe of discourse X has a membership degree $\mu_A(x) \in [0, 1]$ to a fuzzy set A. In contrast, classical (crisp) sets define membership using a binary function:

$$\mu_A(x) = \begin{cases} 1, & \text{if } x \in A \\ 0, & \text{if } x \notin A \end{cases} \tag{2.1}$$

Fuzzy sets allow partial inclusion, which makes them particularly suitable for modeling real-world categories with blurred boundaries. The semantic richness of fuzzy sets supports reasoning with linguistic quantifiers such as “most,” “few,” or “approximately 0.8,” which are not naturally handled in classical logic systems (Liberatore, 2002; Zadeh, 1983).

2.2.2. Core Concepts and Mathematical Principles of Fuzzy Logic

At the heart of fuzzy logic lies the concept of the fuzzy set, which generalizes the classical notion of set membership. A fuzzy set A over a universe X is defined by a membership function:

$$\mu_A : X \to [0, 1] \tag{2.2}$$

where each element $x \in X$ is assigned a degree of membership $\mu_A(x)$ (Beliakov, 1996; Dombi & Csiszár, 2021; Sadollah, 2018). The flexibility of fuzzy sets enables the representation of vague and linguistic concepts with partial truth values.

Membership functions are the cornerstone of fuzzy modeling, as they determine how input values are translated into degrees of membership. Several standard shapes are commonly employed due to their interpretability and computational efficiency. The most widely adopted forms include triangular, trapezoidal, and Gaussian membership functions, each exhibiting distinct characteristics and mathematical formulations, as detailed in the following paragraphs.

The triangular membership function is defined by three parameters $a < b < c$, which determine the triangle's base and peak:

$$\mu(x) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{b-a}, & a < x \leq b \\ \frac{c-x}{c-b}, & b < x \leq c \\ 0, & x > c \end{cases} \tag{2.3}$$

This function increases linearly from a to b, reaches its peak at b, and then decreases linearly to 0 at c.

The trapezoidal membership function generalizes the triangular one using four parameters $a < b \leq c < d$, defining the base and the flat top:

$$\mu(x) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{b-a}, & a < x \leq b \\ 1, & b < x \leq c \\ \frac{d-x}{d-c}, & c < x \leq d \\ 0, & x > d \end{cases} \tag{2.4}$$

It allows representing concepts that are fully true within a range $[b, c]$ and gradually transition outside it.

The Gaussian membership function is defined as:

$$\mu(x) = \exp\left(-\frac{(x-c)^2}{2\sigma^2}\right) \tag{2.5}$$

Here, c is the center (mean) of the distribution and σ controls its spread. This function provides a smooth and differentiable representation of uncertainty.

These functions are selected based on the specific characteristics of the application and can be fine-tuned using expert knowledge, statistical data, or empirical evaluation (Dombi & Csiszár, 2021; Pedrycz & Wang, 2015; Sadollah, 2018). Their mathematical form affects both the precision and the interpretability of the fuzzy system.

The construction of membership functions plays a critical role in the effectiveness of fuzzy models. For instance, triangular and trapezoidal functions are frequently employed in control systems due to their simplicity and ease of implementation, while Gaussian functions offer smoother transitions and are preferred when uncertainty modeling is essential (Dombi & Csiszár, 2021; Sadollah, 2018). Regardless of their shape, all membership functions must satisfy the following basic condition:

$$\mu_A(x) \in [0, 1] \tag{2.6}$$

2.2.3. Fuzzy Reasoning and Inference Rules

Fuzzy logic provides a systematic framework for modeling uncertainty, incorporating approximate reasoning through fuzzy rules and the capacity to integrate predicate logic with probabilistic elements (Mendel, 1995; Zadeh, 1992). This is particularly useful in control and expert systems, where interpretability, gradation, and adaptability are essential.

A canonical example of fuzzy reasoning is expressed in the form of an implication rule such as:

IF temperature IS high **THEN** fan_speed IS fast (2.7)

In this case, both `high` and `fast` correspond to fuzzy sets. Their interaction depends on the degrees of membership assigned to the input variables and the strength of the rule. Unlike binary logic, fuzzy rules can be partially activated, contributing proportionally to the system output.

2.2.4. T-norms, S-norms, and Fuzzy Modifiers

In fuzzy inference systems, reasoning is carried out by combining and modulating the degrees of membership associated with fuzzy sets. This process relies on three core components: triangular norms (*t-norms*), triangular conorms (*s-norms*), and fuzzy modifiers (or linguistic hedges).

A **t-norm** is a binary operator $T : [0, 1]^2 \to [0, 1]$ that generalizes the logical AND by combining fuzzy truth values. Common examples include the minimum operator $T_{\min}(a, b) =$

$\min(a, b)$ and the product $T_{\text{prod}}(a, b) = a \cdot b$, both satisfying properties such as commutativity, associativity, monotonicity, and the identity element $T(a, 1) = a$ (Klement et al., 2013).

Similarly, an **s-norm** $S : [0, 1]^2 \to [0, 1]$ extends the logical OR operation. Examples include the maximum operator $S_{\max}(a, b) = \max(a, b)$ and the probabilistic sum $S_{\text{prob}}(a, b) = a + b - ab$, with neutral element 0 (i.e., $S(a, 0) = a$).

Fuzzy modifiers are unary functions $M : [0, 1] \to [0, 1]$ that alter membership degrees to reflect linguistic nuances. For instance, the intensifier "very" can be represented as $M_{\text{very}}(a) = a^2$, and the negation "not" as $M_{\text{not}}(a) = 1 - a$ (Dombi & Csiszár, 2021).

These operators complete the foundational elements of fuzzy inference and provide formal mechanisms for rule combination and semantic modulation.

2.3. Fuzzy Control Systems (FCS)

Fuzzy control systems (FCS) are rule-based mechanisms designed to emulate human reasoning in decision-making and control processes. They operate by evaluating fuzzy logic rules over input variables, employing membership functions to represent linguistic terms, and generating output decisions in a flexible and interpretable manner. FCS are particularly effective in environments characterized by uncertainty, nonlinearity, or incomplete system knowledge (Arun & Mohan, 2017; Kumari & Mohan, 2024; Praharaj & Mohan, 2022).

2.3.1. Structure of Mamdani-Type Controllers

Among the various types of fuzzy inference systems, the Mamdani-type controller—originally introduced by (Procyk & Mamdani, 1979)—is the most widely adopted due to its intuitive structure and transparency. These features make it particularly suitable for applications that require human interpretability. The general architecture of a Mamdani FCS comprises three sequential stages: fuzzification, inference, and defuzzification, each governed by a well-defined mathematical formulation (Arun & Mohan, 2018; Kumari & Mohan, 2024), as illustrated in Figure 2.3.

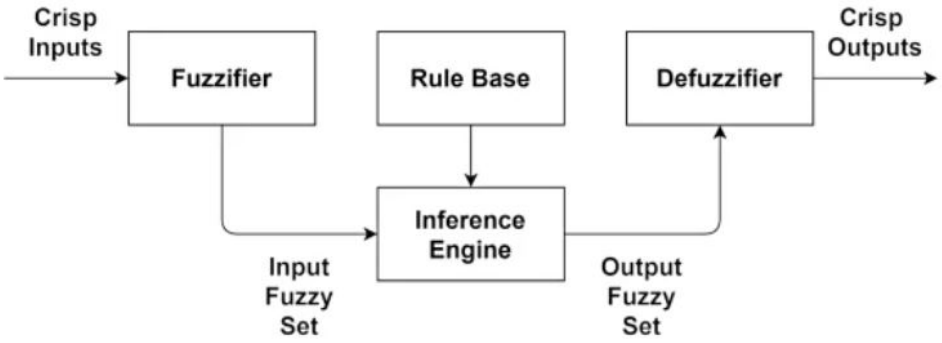


Figure 2.3: Mamdani Fuzzy Control System.

Fuzzification. This stage transforms crisp numerical inputs into fuzzy values by computing their degree of membership to predefined fuzzy sets. Given an input x and a fuzzy set A, the fuzzification process evaluates the membership function $\mu_A(x) \in [0, 1]$. This value quantifies the extent to which the input corresponds to a fuzzy concept (e.g., `high temperature`). The specific membership function $\mu_A(x)$ is selected according to the nature of the input variable and may adopt triangular, trapezoidal, or Gaussian shapes, as previously discussed (Pedrycz & Wang, 2015; Sadollah, 2018).

Inference Engine (Rule Evaluation and Aggregation). The inference stage applies a rule base composed of fuzzy **IF-THEN** statements. A typical fuzzy rule takes the form:

$$\textbf{IF } x_1 \text{ is } A_1 \textbf{ AND } x_2 \text{ is } A_2 \textbf{ THEN } y \text{ is } B \tag{2.8}$$

For each rule, a *firing strength* α is computed to assess the degree to which the antecedents are satisfied. This is typically done using a t-norm operator. One common choice is the minimum function:

$$\alpha = \min\left(\mu_{A_1}(x_1), \mu_{A_2}(x_2)\right) \tag{2.9}$$

The consequent fuzzy set B is then scaled (i.e., truncated) by the firing strength α. All resulting fuzzy outputs from multiple rules are subsequently aggregated using an s-norm operator, often the maximum, to produce a single fuzzy output set (Arun & Mohan, 2017; Kumari & Mohan, 2024; Praharaj & Mohan, 2022).

Defuzzification. The final step in a fuzzy control system is defuzzification, which converts the aggregated fuzzy output set into a single crisp value suitable for control actions. The most commonly used defuzzification technique is the *centroid method* (also known as the center of gravity), defined as:

$$y^* = \frac{\int y \cdot \mu_{\mathrm{B_{agg}}}(y)\, dy}{\int \mu_{\mathrm{B_{agg}}}(y)\, dy} \tag{2.10}$$

Here, y^* denotes the defuzzified crisp output value, and $\mu_{\mathrm{B_{agg}}}(y)$ is the membership function of the aggregated fuzzy output set. Other defuzzification methods include the center of sums, center of area, and mean of maxima, which may be applied depending on the specific requirements of the application (Arun & Mohan, 2018; Praharaj & Mohan, 2022) .

2.3.2. Applications of FCS in Metaheuristics

Fuzzy control systems (FCS) have demonstrated significant utility in the domain of optimization, particularly when embedded within metaheuristic algorithms. Their main advantage lies in providing adaptive parameter tuning mechanisms that respond dynamically to the behavior

of the search process. This capability addresses one of the key limitations of metaheuristics: their performance sensitivity to fixed control parameters (Díaz-Cortés et al., 2017; Valdez et al., 2014).

Metaheuristic algorithms such as PSO, GA, and Differential Evolution (DE) rely on internal parameters, like inertia weights, crossover rates, or mutation factors, that critically influence the balance between exploration and exploitation. Traditionally, these parameters are tuned offline or remain static during execution, potentially resulting in suboptimal convergence or premature stagnation (Sadollah, 2018; Valdez et al., 2010).

By integrating fuzzy controllers, these algorithms can adapt parameters control, adjusting them during the search based on process metrics. For example, in fuzzy-adaptive PSO variants, the inertia weight (w) may be modified dynamically using linguistic rules derived from fuzzy inputs such as `diversity is low` or `fitness improvement is moderate` (Valdez et al., 2014). This allows the algorithm to promote exploration when needed or to intensify exploitation when convergence begins.

Fuzzy systems contribute a human-interpretable reasoning layer to metaheuristics, where expert knowledge or empirical observations are encoded as rule bases. A generic fuzzy rule for parameter adaptation might be IF improvement IS low AND diversity IS low THEN increase inertia weight.

In this context, the fuzzy inputs (e.g., improvement, diversity) are normalized metrics computed from the algorithm's search state, while the outputs are control actions such as adjusting PSO's inertia weight or GA's mutation rate. These outputs are defuzzified and applied to the metaheuristic at each iteration, creating a feedback loop that enhances responsiveness and robustness (Díaz-Cortés et al., 2017; Valdez et al., 2010).

The use of fuzzy control systems has been validated across various problem domains, including engineering design, scheduling, and feature selection, demonstrating improvements in convergence speed, solution quality, and robustness to problem scale and complexity (Sadollah, 2018; Valdez et al., 2014). Notably, the Mamdani-type fuzzy system remains a preferred architecture due to its interpretability and suitability for real-time adjustments.

2.3.3. Limitations of Traditional FCS for Parameter Control

Fuzzy control systems (FCS) have long been recognized as a valuable approach for adaptive parameter control in metaheuristic algorithms due to their ability to manage imprecision and dynamically adjust system behavior. However, traditional implementations—particularly those based on type-1 fuzzy logic—exhibit significant limitations that restrict their effectiveness in dynamic and complex optimization scenarios.

One of the most critical limitations is their reduced capacity to handle uncertainty. Since type-1 fuzzy systems use fixed membership functions, they are often unable to capture the inherent variability and noise present in real-world optimization problems. As a result, their adaptability to evolving search conditions becomes limited, particularly in high-dimensional or stochastic environments (Castillo & Amador-Angulo, 2018; Castillo et al., 2019, 2021; Cuevas

et al., 2022; Patel & Shah, 2022a, 2022b).

Another challenge lies in the computational complexity associated with more advanced fuzzy models. While type-2 and generalized type-2 fuzzy systems offer improved uncertainty handling, they also impose significant computational overhead. This can hinder their practical applicability in time-constrained or resource-limited settings, such as embedded systems or real-time optimization tasks (Castillo et al., 2019, 2021; Patel & Shah, 2022a).

In addition, traditional fuzzy systems face scalability issues as the number of parameters or decision rules increases. Designing a robust and interpretable rule base typically requires extensive expert knowledge and becomes increasingly difficult as metaheuristics grow in complexity. This problem is exacerbated when fuzzy systems are applied across multiple domains or algorithm variants (Jackson et al., 2016; Marques & Gomide, 2011).

Finally, there is evidence that traditional fuzzy controllers may not always outperform simpler heuristic or crisp rule-based approaches. In certain general-purpose applications, the added complexity of fuzzy logic does not consistently translate into better performance, leading to performance plateaus or marginal gains (Jackson et al., 2016).

In response to these limitations, recent studies have explored enhancements such as type-2 fuzzy sets (Castillo & Amador-Angulo, 2018; Patel & Shah, 2022a), interval and shadowed fuzzy representations (Castillo et al., 2019, 2021), and hybrid models that incorporate evolutionary techniques to automate rule design and reduce human dependency (Castillo & Amador-Angulo, 2018; Marques & Gomide, 2011). However, despite these improvements, there remains a gap in methods capable of dynamically adjusting the semantic structure of the fuzzy system itself during execution—a challenge that motivates the adaptive architecture proposed in this thesis.

2.4. Proposed Semantic Adaptation via Fuzzy Schemes

2.4.1. Definition and Formalization of Fuzzy Schemes

Although various authors have explored the design and configuration of fuzzy partitions and sets of membership functions (Dombi & Csiszár, 2021; Sadollah, 2018), the notion of a fuzzy scheme is not formally established as a standard term in the literature, as an example. In this proposal, we define a fuzzy scheme as a structured and coherent configuration of membership functions that determines a semantic partition of a linguistic variable's domain. This operational construct, introduced by the authors, serves as a core abstraction to enable dynamic representation and adaptation of input semantics within fuzzy control systems. Each fuzzy scheme determines how crisp input values are interpreted semantically through a set of overlapping fuzzy sets. In practice, it represents one possible semantic encoding of a variable, and different schemes can lead to significantly different interpretations and control behavior. This interpretive flexibility forms the basis for the semantic adaptation mechanism developed in the present project. Formally, a fuzzy partition $\mathcal{P}$ of a universe X is defined as a finite collection of fuzzy sets $A_1, A_2, \ldots, A_n$, where each set A_i is characterized by a membership function $\mu_{A_i}(x)$. The partition must satisfy the following soft constraint:

$$\sum_{i=1}^{n} \mu_{A_i}(x) \approx 1, \quad \forall x \in X \tag{2.11}$$

This condition ensures that every value x belongs to the fuzzy partition to some degree.

2.4.2. Semantic Modulation in Fuzzy Systems

A fuzzy scheme, as defined here, is not merely a set of isolated fuzzy sets, but an interrelated structure defined by several design elements:

- The number of fuzzy sets composing the partition (e.g., three for `low`, `medium`, `high` versus five for finer granularity);
- The type of membership functions employed (e.g., triangular, trapezoidal, Gaussian);
- The distribution and overlap of the sets across the variable's domain;
- The semantic labels assigned to each fuzzy set (e.g., `low temperature`, `very low temperature`).

These elements collectively determine the resolution, sensitivity, and semantic granularity of the fuzzy controller. For instance, a scheme with three broad triangular sets may support smoother but coarser control decisions, while a five-set Gaussian scheme may capture more nuanced variations at the expense of computational complexity (Pedrycz & Wang, 2015; Sadollah, 2018).

Fuzzy schemes play a pivotal role in shaping how a fuzzy control system interprets and responds to inputs. Since the rule base of a Mamdani controller operates over linguistic values, changing the fuzzy scheme alters the underlying semantics of those values without modifying the rule structure. This introduces a powerful mechanism for semantic modulation, enabling the same control logic to behave differently depending on the active scheme (Dombi & Csiszár, 2021).

In conventional fuzzy systems, the scheme is fixed at design time. However, in adaptive frameworks such as the one proposed here, the controller is capable of selecting among predefined fuzzy schemes at runtime. This selection is guided by search process metrics. By enabling the controller to alter its own semantic representation of the input space, the system acquires a second degree of adaptiveness, enhancing its ability to respond effectively to different phases of the optimization process (Díaz-Cortés et al., 2017; Valdez et al., 2014).

Thus, fuzzy schemes are treated as modular, dynamically interchangeable semantic profiles. Each one defines a coherent fuzzy partition over an input variable, enabling the controller to operate under alternative interpretations of the search landscape. This capacity to switch online between schemes constitutes a form of semantic adaptation, in which the meaning assigned to input variables is no longer fixed but instead context-dependent and adjustable. Unlike traditional approaches that only modify parameter values within a static partition, our proposed system incorporates a higher-level mechanism that reconfigures the semantic structure of the

controller itself. As noted in the literature, such semantic-level adaptation enhances the system's robustness, responsiveness, and ability to generalize across dynamic environments (Castillo & Amador-Angulo, 2018; X. Wang et al., 2023).

2.4.3. Interaction with Static Rule Bases in Mamdani Controllers

In Mamdani-type fuzzy control systems, the core reasoning mechanism relies on a fixed set of fuzzy rules that operate over linguistic variables defined by fuzzy sets. These fuzzy sets are instantiated through a specific fuzzy scheme, understood as a particular configuration of membership functions that defines the semantic partition of the input space. Consequently, the fuzzy scheme plays a critical role in shaping the behavior of the controller, as it determines how numerical inputs are interpreted before rule evaluation.

While the rule base in a Mamdani controller remains static, the meaning of the linguistic terms in both antecedents and consequents is governed by the active fuzzy scheme. For example, the term high diversity may refer to different numerical intervals depending on whether it is defined using triangular or Gaussian functions, and whether the domain is partitioned into three or five fuzzy sets. In this sense, the fuzzy scheme functions as a semantic interface between numerical inputs and the controller's inference layer (Dombi & Csiszár, 2021). This separation between rule structure and membership function configuration is particularly relevant in adaptive control contexts. It allows the controller to adjust its semantic interpretation of inputs without modifying the underlying logic of the rule base. By switching from one fuzzy scheme to another, the system may adopt a more coarse or fine-grained view of the same variable, thus modulating the generalization or specificity of its response (Pedrycz & Wang, 2015; Sadollah, 2018).

In conventional Mamdani systems, the fuzzy scheme is chosen at design time and remains fixed throughout execution. In contrast, the adaptive architecture proposed in this project incorporates a fuzzy controller capable of dynamically selecting among multiple predefined fuzzy schemes during runtime. This semantic adaptation is guided by search process metrics such as stagnation, loss of diversity, or improvement rate (Díaz-Cortés et al., 2017; Valdez et al., 2014). For instance, during early iterations of a metaheuristic, a fuzzy scheme composed of broad, overlapping membership functions may promote exploration by enabling more general interpretations of input conditions. Later in the process, a scheme with narrower and sharper functions can provide more selective control, favoring exploitation. This switching mechanism allows the controller to reconfigure its perception of the input space according to the phase of the optimization process, while keeping the rule base unchanged. The integration of a fuzzy scheme selection module introduces a dual form of adaptivity. First, it supports metaheuristics parameter-level settings by adjusting control variables such as the inertia weight in PSO or the mutation rate in GA. Second, it enables semantic flexibility by altering the internal representation of linguistic variables in response to the observed search behavior. By combining these two mechanisms, the system achieves behavioral plasticity: the capacity to modulate its control behavior based on runtime environmental feedback. This dual adaptation—at both numerical and semantic levels—constitutes a methodological advancement over traditional fuzzy controllers with static partitions and fixed interpretations (Aliev et al., 2025; X. Wang et al., 2023; Xiao et al., 2019).

2.5. Dynamic Scheme Selection Mechanisms

2.5.1. Motivation for Dynamic Semantic Adaptation

The capacity to dynamically adapt fuzzy controllers during execution has been extensively studied, particularly in the context of intelligent control and optimization. Most prior work focuses on the dynamic tuning of fuzzy rule bases, membership functions, or control parameters using external mechanisms. However, to the best of our knowledge, the specific notion of fuzzy schemes, as defined in this proposal as coherent configurations of membership functions forming a fuzzy partition, has not been previously addressed as a modular and dynamically selectable component. This distinction positions the present work within a novel direction that expands the scope of semantic adaptation in fuzzy systems.

Dynamic fuzzy adaptation has demonstrated clear benefits in various domains, including enhanced convergence, improved tracking, reduced steady-state error, and robustness under changing conditions. Early work by Ref. (Hirakawa et al., 2002) introduced adaptive selection of fuzzy rules via automatic tuning of membership functions. Ref (C.-T. Lin & Lee, 1991) proposed neuro-fuzzy systems capable of online rule adaptation, while Ref. (L.-X. Wang & Mendel, 1992) developed hybrid methods to generate and update fuzzy rules dynamically.

The conceptual novelty of this proposal lies in treating fuzzy schemes as first-class adaptive units—each scheme encoding a distinct semantic representation of an input variable through a predefined partition of the domain. Unlike adaptive adjustments of individual membership functions, which may result in uncontrolled semantic drift or inconsistency, the use of predefined, coherent schemes ensures internal consistency and interpretability. Moreover, enabling a controller to switch between schemes allows for dynamic reinterpretation of the same input variable, aligning system behavior with evolving search contexts. This aligns with the broader trend in intelligent systems toward modularity, interpretability, and runtime flexibility. From a cognitive systems perspective, it can be likened to switching between different "modes of perception" depending on environmental cues, supporting the design of more context-aware metaheuristic adaptation mechanisms.

2.5.2. Available Techniques for dynamic selection

Multiple techniques have been proposed for implementing dynamic fuzzy adaptation. These include: Stochastic methods, such as evolutionary strategies and genetic algorithms, used to evolve rule bases or membership function parameters during runtime (Haighton et al., 2024; Qiu et al., 2020; Zhu et al., 2020).

- Reinforcement learning, in particular Q-learning and actor-critic models, which allow fuzzy systems to learn control policies by interacting with their environment and receiving feedback (Haighton et al., 2024; Qiu et al., 2020; Zhu et al., 2020).
- Event-triggered mechanisms, in which structural changes to the fuzzy system are made only when specific conditions or thresholds are met, reducing computational overhead while retaining adaptability (Haighton et al., 2024; Qiu et al., 2020; Si et al., 2022; Zhu et al., 2020).

- Neuro-fuzzy hybrid models, which allow the continuous online adjustment of fuzzy rules or membership parameters based on gradient descent or Hebbian learning (Li & Dong, 2023; Si et al., 2022).

While these methods primarily operate on individual rules or membership functions, they provide a conceptual foundation for this proposal's approach: runtime selection of complete fuzzy schemes. Each scheme represents a self-contained semantic profile over an input variable (metric from the process), allowing the controller to operate under distinct interpretations depending on the search context.

Building upon these techniques, the dynamic selection of fuzzy schemes is framed here as a higher-order control problem. Rather than modifying internal parameters directly, the system must choose among discrete fuzzy configurations that each impose a different bias on the inference process. This design parallels the concept of meta-control in cognitive architectures, where different strategies are selected based on situational awareness.

To enable such dynamic selection, a key challenge is defining selection criteria that reliably indicate when a switch is needed. Candidate criteria include: (i) stagnation of the metaheuristic's objective function; (ii) sudden changes in diversity metrics; (iii) deterioration of exploitation indicators; and (iv) the lack of improvement over a given time window. These signals can trigger scheme switches either deterministically (e.g., threshold rules) or probabilistically (e.g., softmax-based choice), depending on the desired trade-off between stability and reactivity. This framework provides a foundation for implementing lightweight hyperheuristics or reinforcement-learning-based selectors without compromising the interpretability of the fuzzy controller.

3. RESEARCH PROPOSAL

3.1. Motivation and contribution

3.1.1. Motivation and Design Principles

Traditional parameter control mechanisms in metaheuristics—whether fixed, deterministic, or rule-based—often lack the responsiveness required to adapt to the dynamic nature of the search process. Even when fuzzy control systems are incorporated, they frequently rely on fixed membership functions and static rule bases, limiting their ability to reflect evolving optimization contexts. This motivates the development of a control mechanism that can not only adjust parameter values but also revise its own semantic interpretation of the optimization landscape in real-time.

The design of the proposed system is guided by the following principles:

- **Adaptivity:** The control mechanism must dynamically respond to signals such as stagnation, convergence speed, or diversity changes without relying on problem-specific tuning.
- **Semantic Flexibility:** Instead of rigid membership structures, the system should incorporate multiple fuzzy schemes representing alternative interpretations of the same input metrics.
- **Modularity and Generalizability:** The architecture should be applicable across different metaheuristic frameworks, such as PSO and GA, and across problem domains.
- **Transparency and Interpretability:** Although adaptive, the system must retain its interpretability by employing Mamdani-type fuzzy rules, enabling post-hoc analysis and rule inspection.

3.1.2. Core Contribution: Dual-Level Adaptation (Parameters + Semantics)

The core innovation of the proposal lies in its dual-layer adaptation strategy, where both control *parameters* and their semantic representation (*fuzzy schemes*) are adjusted during optimization. This mechanism is structured as follows:

1. **Parameter-Level Adaptation:** At each iteration, fuzzy rules determine the value of a control parameter (e.g., inertia weight in PSO or mutation rate in GA) based on current indicators of search progress.
2. **Semantic-Level Adaptation:** The fuzzy system dynamically selects among a predefined set of fuzzy schemes—each consisting of a coherent set of membership functions. This selection alters the interpretation of linguistic variables (e.g., “low diversity,” “high convergence”) depending on the current search behavior.

This dual adaptation is driven by real-time metrics such as diversity, improvement rate, convergence speed, and stagnation index. Through this architecture, the system not only con-

trols behavior but also redefines how it perceives the current state of the optimization process, enabling context-sensitive reasoning and improved performance across heterogeneous problem types.

3.2. System Architecture and Control Strategy

3.2.1. Functional Architecture Overview

The adaptive fuzzy control system comprises several interconnected components that jointly support both levels of adaptation. These include:

- A metric computation module that quantifies diversity, improvement, and convergence behavior.
- A selector that evaluates conditions for semantic scheme switching, based on predefined criteria or learning policies.
- A Mamdani-type fuzzy inference system whose input sets vary depending on the active fuzzy scheme.
- A parameter update module that injects the adapted control value into the metaheuristic's population update procedure.

This modular architecture facilitates experimentation with different fuzzy schemes, rule sets, and selection strategies, and is designed to be embedded directly into the iterative loop of a metaheuristic algorithm. Full system details, including input/output definitions and flow diagrams, are provided in the following sections.

3.2.2. Inputs, Outputs, and Internal Components

To contextualize the proposed adaptive fuzzy control system, it is instructive to first examine the structure and limitations of conventional fuzzy controllers. As shown in Figure 3.1, the conventional Mamdani Fuzzy Control System operates with a **fixed set of membership functions** and **manually defined rules**. While it can modulate metaheuristic parameters based on input values, its semantic interpretation of those values does not evolve during execution, limiting its responsiveness to different optimization phases.

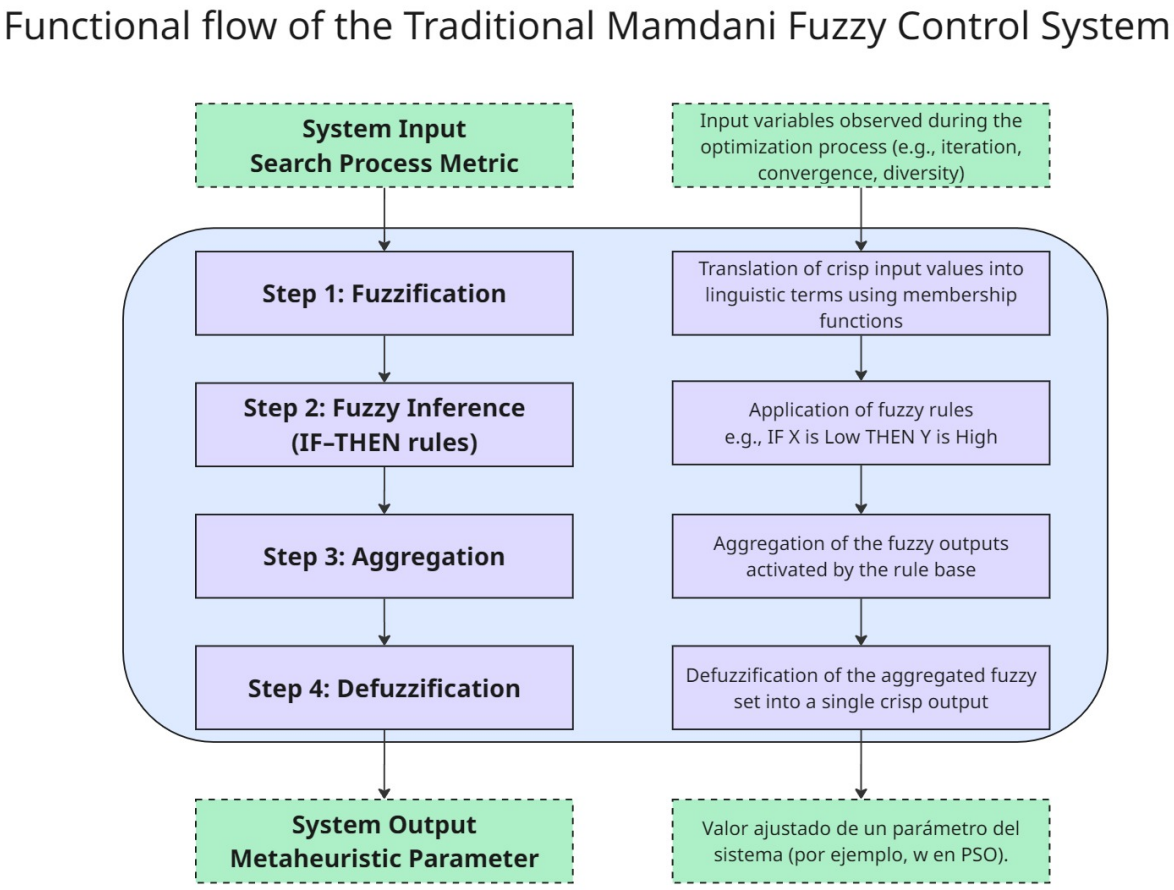


Figure 3.1: Functional flow of the Traditional Mamdani Fuzzy Control System.

In contrast, the proposed architecture introduces a dual-layer adaptation mechanism that enhances the responsiveness and plasticity of the control process. As illustrated in Figure 3.2, the system dynamically selects among a predefined set of fuzzy schemes based on runtime observations of search process metrics. This selection precedes the standard Mamdani inference cycle—fuzzification, fuzzy inference, aggregation, and defuzzification—which remains structurally intact but semantically enriched.

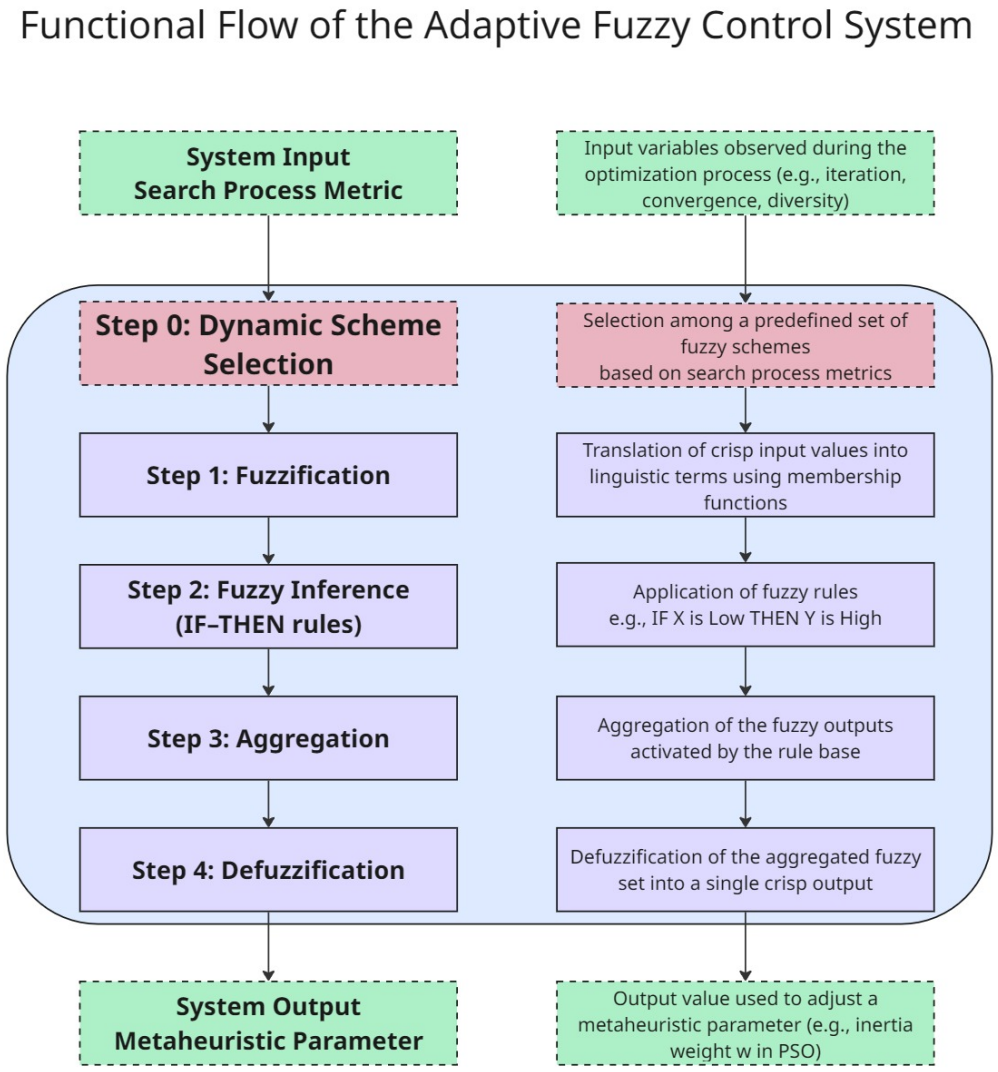


Figure 3.2: Functional flow of the proposed adaptive fuzzy control system.

Each fuzzy scheme defines a distinct configuration of membership functions over the input space, allowing the system to change its semantic interpretation of linguistic variables during execution. The decision to switch schemes is guided by real-time indicators such as diversity, stagnation, or improvement rate, enabling the control system to adjust not only parameter values but also the meaning attributed to search process states.

The internal operation of the adaptive controller is summarized in Algorithm 3.1. This procedure describes how real-time metrics are interpreted through dynamic fuzzy schemes to produce a context-sensitive control parameter.

Algoritmo 5.1 Adaptive Fuzzy Control System (AFCS)

Require: Process metrics $\mathcal{M} = \{\text{diversity}, \text{stagnation}, \text{improvement rate}, \text{iteration index}\}$
Require: Set of fuzzy schemes $\mathcal{S} = \{S_1, S_2, \ldots, S_n\}$, each with associated membership functions
Ensure: Adapted parameter value θ (e.g., inertia weight w, mutation rate μ)

```
Acquire current values of metrics: M_t
Select fuzzy scheme S* ∈ S based on M_t and a predefined selection strategy
for each input metric m ∈ M_t do
    Compute degree of membership μ_m using membership functions in S*
end for
Apply fuzzy inference rules over all μ_m
Aggregate the outputs into a fuzzy set
Perform defuzzification to compute the crisp output θ
return adapted parameter θ
```

3.2.3. Integration with Metaheuristics (PSO and GA)

The proposed adaptive fuzzy control system (AFCS) is designed to function as an embedded decision-making module within population-based metaheuristics. In this study, it is specifically integrated into two representative algorithms: Particle Swarm Optimization (PSO) and Genetic Algorithm (GA). These methods were selected for their contrasting exploration–exploitation dynamics and their proven effectiveness across a wide range of optimization domains. The standard versions of PSO and GA are presented in Algorithms 3.2 and 3.3, respectively.

Algoritmo 5.2 Standard Particle Swarm Optimization (PSO)

```
Initialize swarm of particles with random positions x_i and velocities v_i
for each particle i do
    Evaluate fitness f(x_i)
    Set personal best p_i ← x_i
end for
Set global best g ← arg min f(p_i)
while termination criterion not met do
    for each particle i do
        v_i ← w v_i + c1 r1 (p_i − x_i) + c2 r2 (g − x_i)
        x_i ← x_i + v_i
        Evaluate fitness f(x_i)
        Update p_i and g if necessary
    end for
end while
```

Algoritmo 5.3 Standard Genetic Algorithm (GA)

1: Initialize population P with random individuals
2: Evaluate fitness $f(x)$ for each individual $x \in P$
3: **while** termination criterion not met **do**
4: Select parents from P using selection operator
5: Apply crossover to generate offspring
6: Apply mutation to offspring
7: Evaluate fitness of offspring
8: Form new population P by selecting the best individuals (elitism or replacement)
9: **end while**

Once embedded, the adaptive fuzzy control system (AFCS) continuously receives real-time metrics that characterize the state of the search process—such as diversity, improvement rate, and stagnation level. These indicators are interpreted through a fuzzy scheme that is dynamically selected at each iteration. Based on the fuzzy inference process, the controller generates a continuous-valued control signal, which is used to update a key algorithmic parameter, such as the inertia weight w in PSO or the mutation rate μ in GA. This adaptive mechanism allows the metaheuristic to responsively modulate its search behavior, enhancing its ability to balance exploration and exploitation as the optimization evolves.

The integration of the AFCS is deliberately designed to be modular and non-intrusive. Rather than altering the internal evolutionary operators—such as selection, crossover, or velocity updates—the controller interfaces exclusively with the parameter adaptation stage. At each iteration, it provides a revised parameter value based on the current state of the search, without modifying the metaheuristic's core logic. This design promotes interoperability, enabling seamless incorporation of the AFCS into a wide variety of optimization frameworks.

To illustrate this integration, we extend the standard formulations of Particle Swarm Optimization (PSO) and Genetic Algorithm (GA) by embedding the AFCS as a parameter adaptation module. In PSO, the controller dynamically tunes the inertia weight w, a parameter critical for regulating the exploration–exploitation tradeoff. In GA, it adaptively adjusts the mutation rate μ, promoting population diversity over time. In both cases, the controller operates within the main iteration loop, continuously processing runtime metrics to guide the adaptation process.

The full implementation of these adaptive versions is presented in Algorithm 3.4 and Algorithm 3.5, respectively.

Algoritmo 3.4 Adaptive PSO with Fuzzy Control System (FCS)

Initialize swarm of particles with random positions x_i and velocities v_i
Initialize fuzzy control system with predefined fuzzy schemes
for each particle i **do**
    Evaluate fitness $f(x_i)$
    Set personal best $p_i \leftarrow x_i$
end for
Set global best $g \leftarrow \arg\min f(p_i)$
while termination criterion not met **do**
    Compute search process metrics (e.g., diversity, improvement rate, stagnation)
    Use FCS to select fuzzy scheme and adapt inertia weight w
    for each particle i **do**
        $v_i \leftarrow wv_i + c_1 r_1(p_i - x_i) + c_2 r_2(g - x_i)$
        $x_i \leftarrow x_i + v_i$
        Evaluate fitness $f(x_i)$
        Update p_i and g if necessary
    end for
end while

Algoritmo 3.5 Adaptive GA with Fuzzy Control System (FCS)

Initialize population P with random individuals
Initialize fuzzy control system with predefined fuzzy schemes
Evaluate fitness $f(x)$ for each individual $x \in P$
while termination criterion not met **do**
    Compute search process metrics (e.g., diversity, improvement, stagnation)
    Use FCS to adapt mutation rate μ
    Select parents from P using selection operator
    Apply crossover to generate offspring
    Apply mutation using adapted μ
    Evaluate fitness of offspring
    Form new population P by selecting the best individuals
end while

3.3. Semantic Adaptation and Switching Strategies

3.3.1. Switching Criteria and Process Metrics

To enable context-sensitive behavior, the decision to switch from one fuzzy scheme to another is guided by observable metrics that reflect the state of the optimization process. These metrics include:

- Diversity, which quantifies the structural dispersion of the population in the search space and supports the monitoring of exploration behavior (Castillo et al., 2017; Patel & Shah, 2022b);

- Convergence speed, which measures the degree to which candidate solutions approach local or global optima and reflects exploitation dynamics (Castillo et al., 2017; Patel & Shah, 2022b);
- Improvement rate, which captures the pace of fitness enhancement across iterations, serving as an indicator of search progress (Castillo et al., 2017; Patel & Shah, 2022b);
- Stagnation, defined as a sequence of iterations without improvement, signaling potential entrapment in local optima or lack of movement in the search landscape (Castillo et al., 2017; Patel & Shah, 2022b);
- Iteration Index, which encodes the current stage of the optimization process (scaled between 0 and 1), enabling temporally contextualized control behavior (Peraza et al., 2018).

These metrics are simultaneously interpreted to characterize the current search status and to inform the upper-level selector module responsible for determining whether a change in fuzzy scheme is warranted. By dynamically adjusting the controller's semantic configuration in response to these indicators, the system aims to improve performance across a range of problem scenarios.

To assess the effectiveness of this semantic adaptation, four key performance dimensions will be considered:

1. Effectiveness: Measures the algorithm's ability to find high-quality or optimal solutions and it will be measured by best fitness values obtained, relative error to known optima, and success rate across independent runs (Halim et al., 2021);
2. Efficiency: Refers to the computational resources required and it will be measured via the function evaluations required to reach a target performance, as well as average computational time (Halim et al., 2021);
3. Robustness: quantified through the statistical dispersion of results using standard deviation, interquartile range (Djebedjian et al., 2021).
4. Exploration–Exploitation Balance: measured by the exploration–exploitation ratio (Crawford et al., 2021).

By introducing a mechanism for dynamic scheme selection, this research advances a novel form of semantic-level adaptation. Switching between fuzzy schemes—each encoding a distinct interpretation of the input space—enables the controller to reconfigure both its decision logic and internal semantic representation at runtime. Each scheme functions as a behavioral mode, supporting exploratory, conservative, or balanced dynamics. When combined with adaptive selection strategies, this approach grants the system a degree of semantic flexibility and behavioral plasticity that surpasses conventional fuzzy control systems, making it particularly well-suited for complex or dynamic optimization environments.

3.3.2. Scheme Selection Methods

Depending on the level of adaptability desired, this module may implement:

- Static rule-based switching, triggered by simple logical conditions(Cord et al., 2001);

- Stochastic profile selection, where schemes are chosen probabilistically based on historical performance (Haighton et al., 2024; Qiu et al., 2020; Zhu et al., 2020).
- Reinforcement learning strategies, such as Q-learning, enabling the controller to learn a policy for scheme selection across search contexts (Haighton et al., 2024; Qiu et al., 2020; Zhu et al., 2020).
- Hyperheuristic approaches, where fuzzy schemes are treated as low-level heuristics governed by a higher-level learning algorithm (Burke et al., 2010; Crawford et al., 2012).

Each strategy will be empirically validated using benchmark problems in continuous optimization, combinatorial search, and feature selection.

3.4. Experimental and Validation Plan

To assess the viability of the proposed approach, we introduce an experimental plan that evaluates both the internal behavior and external performance of the system. As described in previous chapters, this adaptive fuzzy control system is embedded within a metaheuristic framework and designed to modulate internal parameters through a dual-layer adaptation mechanism: (i) parameter-level setting and (ii) semantic-level modulation via the dynamic selection of fuzzy schemes. The system operates as a feedback controller that interprets process metrics—such as diversity, convergence speed, improvement rate, stagnation, and iteration index—using fuzzy linguistic variables. Based on these inputs, it dynamically adjusts metaheuristic parameters in PSO and GA, aligning control behavior with the evolving state of the search process.

3.4.1. Benchmark Problems

To assess the performance and generalization capability of the proposed architecture, a comprehensive experimental protocol will be carried out. The system will be benchmarked on a set of representative optimization problems, covering multiple domains:

- Continuous functions from the IEEE Congress on Evolutionary Computation (CEC) benchmark suites;
- Combinatorial problems such as the Set Covering Problem (SCP) and the Knapsack Problem (KP);
- The Feature Selection Problem, characterized by high-dimensional, binary, and sparse search spaces. In this context, we aim to evaluate the system both on synthetic benchmark datasets and on real-world classification scenarios, such as Biomedical or Educational data. This dual perspective will allow us to assess not only the system's optimization capacity but also its practical relevance and transferability to applied learning tasks (Barrera-García et al., 2023; Cisternas-Caneo et al., 2025).

3.4.2. Evaluation Metrics

The system's performance will be assessed in terms of **effectiveness**, **efficiency**, **robustness**, and **exploration–exploitation balance**, using the validation metrics already defined earlier. These criteria will guide a comparative evaluation of the different fuzzy scheme selection strategies

under controlled experimental conditions.

3.4.3. Statistical Testing and Reproducibility

Each experimental condition will be executed multiple times under controlled random seeds to ensure statistical validity. Statistical significance will be tested using non-parametric tests such as the Friedman and Wilcoxon signed-rank procedures, in accordance with standard metaheuristic benchmarking practices. Convergence behavior will also be visually analyzed through fitness trajectory plots.

The entire experimentation pipeline will be implemented in Python, employing libraries such as `SciPy`, `NumPy`, and `scikit-posthocs`. The complete framework (code, instances, results) will be made publicly available online through a dedicated GitHub repository, ensuring transparency, reproducibility, and open access. Given the computational complexity of the proposed protocol, dedicated infrastructure (e.g., high-performance computing nodes or GPU clusters) will be secured to guarantee timely execution.

3.4.4. Expected Outputs and Deliverables

- A modular and reusable fuzzy controller supporting dynamic scheme selection for online metaheuristic parameter setting;
- A comparative evaluation of alternative fuzzy scheme selection mechanisms (static rules, stochastic strategies, reinforcement learning, hyperheuristics);
- A set of benchmarking reports and visualizations documenting performance across diverse problem domains;
- At least six (SCI or SCIE) WoS-indexed journal articles and six Scopus-indexed conference papers reporting on theoretical, methodological, and empirical contributions.

4. METHODOLOGY

4.1. Research Questions

1. Can the dynamic selection of fuzzy schemes during the execution of a metaheuristic algorithm lead to improved optimization performance compared to traditional static fuzzy control configurations?
2. Which dynamic selection mechanism (static, stochastic, reinforcement learning, hyperheuristic) is the most effective for guiding the adaptation of fuzzy schemes in the context of complex optimization problems?
3. What impact does a fuzzy control system optimized by machine learning have on the dynamic tuning of metaheuristic parameters applied to different optimization problems?

4.2. Research Hypothesis

The integration of an adaptive fuzzy control system that dynamically selects among predefined fuzzy schemes for online parameter tuning of metaheuristics, guided by feedback from search process metrics, will significantly enhance the effectiveness, efficiency, robustness, and exploration–exploitation balance of such algorithms compared to approaches using static fuzzy configurations or fixed parameter settings.

4.3. General Objective

To design and evaluate an adaptive fuzzy control system that, within metaheuristic optimization frameworks, is capable of dynamically selecting fuzzy schemes to control algorithm parameters based on real-time search process metrics, in order to improve effectiveness, efficiency, robustness, and exploration–exploitation balance.

Specific Objectives

1. To define a structured set of fuzzy schemes, each representing an alternative semantic configuration of the fuzzy system's input variables, characterized by variation in membership function type, quantity, shape, domain coverage, and degree of overlap.
2. To develop a Mamdani-type fuzzy controller that selects, at runtime, the most appropriate fuzzy scheme based on search process metrics such as diversity, improvement rate, convergence speed, stagnation, and iteration index.
3. To implement and compare multiple selection strategies for fuzzy scheme adaptation, including static rule-based mechanisms, stochastic switching, reinforcement learning (Q-Learning), and hyperheuristic approaches.

4. To integrate the proposed adaptive controller into representative metaheuristics, including PSO and GA, selected for their complementary exploration–exploitation dynamics.
5. To empirically evaluate the proposed system using both benchmark mathematical functions (IEEE CEC) and benchmark of combinatorial problems (Set Covering, Knapsack, and Feature Selection) and real world applications of predictive models of Biomedical and Educational data optimized by Feature Selection, analyzing its performance under controlled experimental conditions according to metrics of effectiveness, efficiency, robustness and and exploration–exploitation balance.

5. WORK PLAN

The project is organized into five Work Packages (WPx) that structure the research and development activities required to implement and validate the proposed adaptive fuzzy control system. Each WP is composed of specific tasks (Tx) and expected achievements (Ax). A detailed timeline is presented in the Work Plan section.

WP1: Definition of fuzzy schemes and semantic representation schemes This work package directly supports Objective 1 by designing and formalizing fuzzy schemes as modular, reusable semantic profiles. These schemes vary in type, resolution, and structure, enabling multiple interpretations of input variables to be represented within the control system.

Tasks:

- T1.1 Survey of fuzzy membership function types and semantic partitioning strategies.
- T1.2 Definition of granular configurations of fuzzy partitions that encode varying semantic resolutions, with emphasis on functional diversity and domain coverage.
- T1.3 Implementation of fuzzy profile structures in a configurable control system.
- T1.4 Design of semantic metrics to compare the effect of each profile on the fuzzy controller's behavior.
- T1.5 Writing of articles to disseminate the theoretical framework and the taxonomy of fuzzy schemes.

Achievements:

- A1.1 Taxonomy of fuzzy schemes /schemes for adaptive fuzzy control.
- A1.2 Software library of predefined fuzzy schemes for integration in metaheuristics.
- A1.3 Submission of at least one (SCI or SCIE) WoS-indexed journal article and one Scopus-indexed conference paper on fuzzy profile design and semantic flexibility.

WP2: Design of the adaptive fuzzy control system This WP supports Objective 2 and involves designing and implementing an adaptive fuzzy control system capable of switching between fuzzy schemes at runtime. The selection will be guided by observed behavior in the search process metrics.

Tasks:

- T2.1 Design of the fuzzy inference system (Mamdani-type) with dynamic profile switching.
- T2.2 Definition of input metrics (a priori: diversity, convergence speed, improvement rate, stagnation, and iteration index).

- T2.3 Implementation of runtime evaluation and switching mechanisms.
- T2.4 Validation of the adaptive system under static and dynamic profile scenarios.
- T2.5 Writing of articles on the architecture and design of the adaptive fuzzy system.

Achievements:

- A2.1 Implementation of a modular adaptive fuzzy control system.
- A2.2 Validation of the dynamic profile switching mechanism.
- A2.3 Submission of at least one (SCI or SCIE) WoS-indexed journal article and one Scopus-indexed conference paper on system architecture and adaptive control behavior.

WP3: Dynamic selection strategies for fuzzy schemes In this WP, directly related to Objective 3, several strategies for fuzzy profile selection will be developed and compared, including static control (baseline), stochastic methods, reinforcement learning (Q-learning), and hyperheuristics. These strategies will govern the semantic adaptation capability of the system by enabling contextual switching between fuzzy schemes according to online search process metrics feedback.

Tasks:

- T3.1 Review of existing dynamic selection techniques.
- T3.2 Implementation of dynamic profile selectors based on different strategies (a priori stochastic methods, Q-learning, and hyperheuristics).
- T3.3 Comparative analysis of selector performance in offline simulations.
- T3.4 Feedback-based refinement of selection strategies.
- T3.5 Writing of articles on the comparative performance of selection strategies.

Achievements:

- A3.1 Software components for profile selection using different strategies.
- A3.2 Benchmarking results comparing static and dynamic approaches.
- A3.3 Submission of at least two (SCI or SCIE) WoS-indexed journal articles and two Scopus-indexed conference papers comparing the methodological strengths of static, stochastic, and learning-based selection strategies for semantic control.

WP4: Integration, testing, and benchmarking in metaheuristics Directly supporting Objectives 4 and 5, this work package operationalizes the adaptive fuzzy control system by integrating it into state-of-the-art metaheuristics and validating its impact on a suite of classical and real-world optimization problems. The experimental evaluation focuses on quantifying the effectiveness, efficiency, robustness, and exploration–exploitation balance of different dynamic selection strategies within PSO and GA frameworks.

Tasks:

- T4.1 Selection and preprocessing of benchmark problems, including mathematical functions from the IEEE Congress on Evolutionary Computation (CEC), combinatorial problems such as the Set Covering Problem (SCP) and the Knapsack Problem (KP), and the Feature Selection Problem.
- T4.2 Integration of the adaptive fuzzy control system into PSO and GA implementations.
- T4.3 Design and execution of comparative experimental protocols under controlled settings, including static and dynamic fuzzy configurations.
- T4.4 Analysis of the impact of dynamic fuzzy scheme selection in terms of optimization process indicators.
- T4.5 Application of the design system on real-world cases of predictive models of Biomedical and Educational data optimized by Feature Selection, reducing the dimensionality problem.
- T4.6 Writing and submission of scientific articles reporting the results and methodological contributions from the application of the system to each problem domain.

Achievements:

- A4.1 Fully integrated and operational adaptive fuzzy control modules within PSO and GA.
- A4.2 Comprehensive benchmarking dataset and experimental framework for evaluating fuzzy scheme selection strategies.
- A4.3 Design and execution of comparative experimental protocols under controlled settings, evaluating static fuzzy configurations as baseline and all dynamic scheme selection strategies.
- A4.4 Analytical report on the influence of fuzzy scheme semantic characteristics (e.g., number, shape, overlap of membership functions) on exploration–exploitation behavior and performance metrics.
- A4.5 Submission of at least two (SCI or SCIE) WoS-indexed journal articles and two Scopus-indexed conference papers focused on applied optimization results and the generalizability of the proposed system across diverse problem domains.

WP5: Project management and dissemination This work package ensures the effective coordination, documentation, and scientific dissemination of the project's progress and outcomes. It encompasses internal management tasks, communication of intermediate and final results, and engagement with the research community through conferences and peer-reviewed publications.

Tasks:

- T5.1 Development and continuous updating of a project website serving as a public repository for documentation, open-source tools, fuzzy schemes, and benchmarking results.

- T5.2 Coordination and monitoring of research activities, including scheduling, milestones, deliverables, and risk mitigation strategies.
- T5.3 Preparation and submission of interim and final reports to accordance with the project's management requirements.
- T5.4 Dissemination of results through oral and poster presentations at national and international scientific events.
- T5.5 Preparation and submission of at least four peer-reviewed publications, targeting journals and conferences relevant to fuzzy systems, metaheuristics, and intelligent optimization.

Achievements:

- A5.1 Operational project website and repository with publicly accessible code, fuzzy schemes, experimental datasets, and technical documentation.
- A5.2 Timely delivery of all mandatory program reports.
- A5.3 Scientific dissemination plan fulfilled through the submission of a minimum of six (SCI or SCIE) WoS-indexed journal articles and six Scopus-indexed conference papers.

5.0.1. Gantt chart

The Gantt chart (Fig. 5.1) has been completely designed according to the stages proposed in the methodology. For details of the activities of each stage, please refer to the methodology section.

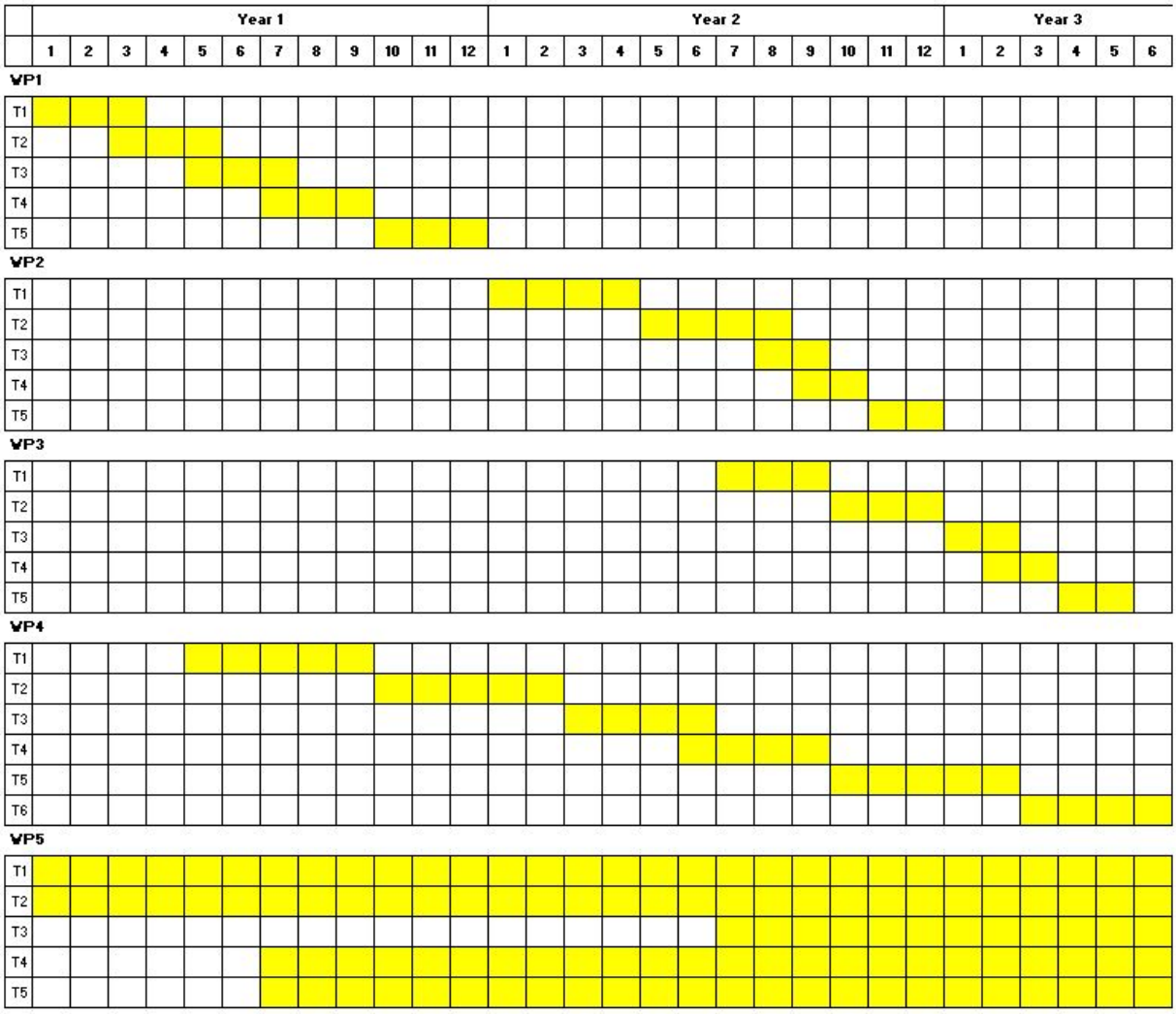


Figure 5.1: Gantt Chart

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